

Laboratory work 4

Task1:

The screenshot shows a PostgreSQL client interface. On the left, a tree view displays the database structure, with 'laboratory_work_2' selected under 'Databases (5)'. The main panel shows a SQL query in the 'Query' tab:

```
1 SELECT UPPER(airline_name) AS airline_name
2 FROM airline_info;
```

Below the query editor, the 'Data Output' tab is active, displaying the results of the query in a table format. The table has one column, 'airline_name', and six rows of data.

	airline_name
1	AIRLINE_2
2	AIRLINE_3
3	AIRLINE_4
4	AIRLINE_6
5	AIRLINE_7
6	AIRLINE_8

Task2:

The screenshot shows the PostgreSQL IDE interface. On the left, the 'Object Explorer' pane displays the database structure for 'laboratory_work_2', including Casts, Catalogs, Event Triggers, Extensions, Foreign Data Wrappers, Languages, Publications, Schemas (1), Subscriptions, phonebook, phonebook_db, postgres, snake, Login/Group Roles, and Tablespaces. The main query editor displays the following SQL script:

```

1  SELECT UPPER(airline_name) AS airline_name
2  FROM airline_info;
3
4  SELECT REPLACE(airline_name, 'Air', 'Aero') AS airline_name
5  FROM airline_info;
6
7  SELECT airline_name
8  FROM airline_info;
9
10 SELECT REPLACE(airline_name, 'AIR', 'AERO') AS airline_name
11 FROM airline_info;

```

Below the query editor, the 'Data Output' pane shows the results of the first query, displaying a list of airline names in uppercase:

airline_name
AEROLINE_2
AEROLINE_3
AEROLINE_4
AEROLINE_6
AEROLINE_7
AEROLINE_8
AEROLINE_9
AEROLINE_10
AEROLINE_11
AEROLINE_12
AEROLINE_13

Task3:

The screenshot shows the PostgreSQL IDE interface. On the left, the 'Object Explorer' pane displays the database structure for 'laboratory_work_2', including booking_id integer, flight_id integer, passenger_id integer, booking_platform character varying(50), created_at timestamp without time zone, updated_at timestamp without time zone, status character varying(50), ticket_price numeric(7,2), constraints, indexes, ILS Policies, rules, triggers, king_flight, its, columns (12), flight_id integer, sch_departure_time timestamp without time zone, sch_arrival_time timestamp without time zone, departing_airport_id integer, arriving_airport_id integer, departing_gate text, arriving_gate character varying(50), airline_id integer, act_departure_time timestamp without time zone, act_arrival_time timestamp without time zone, created_at timestamp without time zone, updated_at timestamp without time zone, constraints, and indexes. The main query editor displays the following SQL script:

```

88  VALUES (
89      31,
90      '2025-02-01 08:00:00',
91      '2025-02-01 12:00:00',
92      1,
93      2,
94      'Gate_1',
95      'Gate_2',
96      1,
97      '2025-02-01 08:15:00',
98      '2025-02-01 12:15:00',
99      CURRENT_TIMESTAMP,
100     CURRENT_TIMESTAMP
101 );
102
103 SELECT flight_id
104 FROM flights
105 WHERE airline_id IN (1, 2);

```

Below the query editor, the 'Data Output' pane shows the results of the second query, displaying a list of flight IDs:

flight_id
12
28
31

Task4:

The screenshot displays a PostgreSQL IDE interface with the following components:

- Object Explorer (Left Panel):** Shows a hierarchical view of the database schema. The 'airport' table is selected, showing its columns: `airport_id` (integer), `airport_name` (character varying(50)), `country` (character varying(50)), `state` (character varying(50)), `city` (character varying(50)), `created_at` (timestamp without time zone), and `updated_at` (timestamp without time zone). Other tables like `airline`, `baggage`, `baggage_check`, and `boarding_pass` are also visible.
- Query Editor (Center):** Contains a SQL query that inserts a new record into the `airport` table and then selects it. The query is as follows:

```
120      city,  
121      created_at,  
122      updated_at  
123    )  
124    VALUES (  
125      31,  
126      'Reginal Air Airport',  
127      'Kazakhstan',  
128      'Almaty',  
129      'Almaty',  
130      CURRENT_TIMESTAMP,  
131      CURRENT_TIMESTAMP  
132    );  
133  
134    SELECT airport_name  
135    FROM airport  
136    WHERE airport_name LIKE '%Reginal%'  
137    AND airport_name LIKE '%Air%';
```
- Data Output (Bottom):** Shows the result of the query. It displays a single row with the value 'Reginal Air Airport' under the column `airport_name`. The column is identified as `airport_name character varying (50)`.

Task5:

The screenshot displays a PostgreSQL IDE interface. On the left, the 'Object Explorer' shows a database schema with tables including 'airline', 'airport', 'baggage', 'baggage_check', 'boarding_pass', 'booking', 'booking_flight', 'flights', 'passengers', 'receipt_airline', 'receipt_airport', 'receipt_booking', 'receipt_booking_seat', 'receipt_flight', 'receipt_passenger', 'security_check', and 'Trigger Functions'. The 'passengers' table is expanded, showing columns like 'passenger_id', 'first_name', 'last_name', 'date_of_birth', 'gender', 'country_of_citizenship', 'country_of_residence', 'passport_number', 'created_at', and 'updated_at'.

The main query editor shows the following SQL code:

```

125     31,
126     'Reginal Air Airport',
127     'Kazakhstan',
128     'Almaty',
129     'Almaty',
130     CURRENT_TIMESTAMP,
131     CURRENT_TIMESTAMP
132 );
133
134 SELECT airport_name
135 FROM airport
136 WHERE airport_name LIKE '%Reginal%'
137        AND airport_name LIKE '%Air%';
138
139 SELECT first_name,
140        last_name,
141        TO_CHAR(date_of_birth, 'Month DD, YYYY') AS date_of_birth
142 FROM passengers;

```

The 'Data Output' pane shows the results of the query, displaying 12 rows of data. The columns are 'first_name', 'last_name', and 'date_of_birth'.

	first_name	last_name	date_of_birth
1	Alex	Smith	May 01, 1...
2	Adina	Sultanov	April 27, 20...
3	Madina	Omarov	June 28, 1...
4	Timur	Akhmetova	March 29, 2...
5	Aruzhan	Brown	November 18...
6	Dana	Kim	July 09, 20...
7	Alex	Akhmetova	February 26, ...
8	Aliya	Akhmetova	September 02...
9	Diana	Kassenova	June 16, 1...
10	Adina	Sadykova	September 16...
11	Timur	Akhmetova	April 14, 19...
12	Aruzhan	Lee	December 16...

At the bottom, the status bar indicates 'Total rows: 50' and 'Query complete 00:00:00.075'.

Task6:

The screenshot displays a PostgreSQL IDE interface with the following components:

- Object Explorer (Left):** Shows a database schema with tables including `airline`, `airport`, `baggage`, `baggage_check`, `boarding_pass`, `booking`, `booking_flight`, `flights`, `passengers`, `receipt_airline`, `receipt_airport`, `receipt_booking`, `receipt_booking_seat`, `receipt_flight`, `receipt_passenger`, `security_check`, and `Trainer Functions`. The `passengers` table is selected, showing its columns: `passenger_id` (integer), `first_name` (character varying(50)), `last_name` (character varying(50)), `date_of_birth` (date), `gender` (character varying(50)), `country_of_citizenship` (character varying), `country_of_residence` (character varying), `passport_number` (character varying(20)), `created_at` (timestamp without time zone), and `updated_at` (timestamp without time zone).
- Query Editor (Center):** Contains a SQL query with line numbers 130 to 147. The query includes a `CURRENT_TIMESTAMP` statement, a `SELECT` statement for `airport_name` from the `airport` table with filters for `Reginal` and `Air`, and another `SELECT` statement for `first_name`, `last_name`, and `date_of_birth` from the `passengers` table.
- Data Output (Bottom):** Shows the results of the query. The first column is `flight_id` (PK) integer, and the second column is `passenger_id` (PK) integer. The results are a list of 12 rows, numbered 1 to 12, showing the relationship between flight and passenger IDs.
- Status Bar (Bottom):** Indicates "Total rows: 31" and "Query complete 00:00:00.075". A green checkmark icon and the text "Successful" are visible on the right.

Task7:

The screenshot shows a PostgreSQL query editor interface. The query is as follows:

```
136 WHERE airport_name LIKE '%Reginal%'
137 AND airport_name LIKE '%Air%';
138
139 SELECT first_name,
140        last_name,
141        TO_CHAR(date_of_birth, 'Month DD, YYYY') AS date_of_birth
142 FROM passengers;
143
144
145 SELECT flight_id
146 FROM flights
147 WHERE act_arrival_time > sch_arrival_time;
148
149
150 SELECT flight_id,
151        act_arrival_time - sch_arrival_time AS delay_time
152 FROM flights
153 WHERE act_arrival_time > sch_arrival_time;
```

The results of the query are displayed in a table with the following columns: **flight_id** (PK integer) and **delay_time** (interval). The table contains 12 rows of data, all showing a delay of 00:15:00.

flight_id	delay_time
1	00:15:00
2	00:15:00
3	00:15:00
4	00:15:00
5	00:15:00
6	00:15:00
7	00:15:00
8	00:15:00
9	00:15:00
10	00:15:00
11	00:15:00
12	00:15:00

A green success message is visible in the bottom right corner: **✓ Success**. The status bar at the bottom indicates "Total rows: 11" and "Query complete 00:00:00.050".

Task8:

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```
142 FROM passengers;
143
144
145 SELECT flight_id
146 FROM flights
147 WHERE act_arrival_time > sch_arrival_time;
148
149
150 SELECT flight_id,
151        act_arrival_time - sch_arrival_time AS delay_time
152 FROM flights
153 WHERE act_arrival_time > sch_arrival_time;
154
155
156 SELECT *
157 FROM airline_info
158 WHERE airline_country IN ('France', 'Portugal', 'Poland')
159 AND created_at BETWEEN '2023-11-01' AND '2024-03-31';
```

Data Output Messages Notifications

airline_id	airline_code	airline_name	airline_country	created_at	updated_at
[PK] integer	character varying (30)	character varying (50)	character varying (50)	timestamp without time zone	timestamp without time zone

Task9:

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```
148
149
150 SELECT flight_id,
151        act_arrival_time - sch_arrival_time AS delay_time
152 FROM flights
153 WHERE act_arrival_time > sch_arrival_time;
154
155
156 SELECT *
157 FROM airline_info
158 WHERE airline_country IN ('France', 'Portugal', 'Poland')
159 AND created_at BETWEEN '2023-11-01' AND '2024-03-31';
160
161
162 SELECT * FROM baggage
163 WHERE weight_in_kg > 25
164 ORDER BY weight_in_kg DESC
165 LIMIT 3;
```

Data Output Messages Notifications

baggage_id	weight_in_kg	created_at	updated_at	booking_id
[PK] integer	numeric (4,2)	timestamp without time zone	timestamp without time zone	integer

Task10:

The screenshot shows a PostgreSQL IDE interface. On the left, the 'Object Explorer' pane displays a tree structure of database objects. The 'baggage' table is selected, showing its columns: baggage_id (integer), weight_in_kg (numeric(4,2)), created_at (timestamp without time zone), updated_at (timestamp without time zone), and booking_id (integer). The main query editor displays a SQL query with line numbers 157 to 174. The query consists of two parts: a query on the 'airline_info' table and a query on the 'baggage' table. The results pane at the bottom shows the output of the second query, displaying the first and last names of passengers.

```

157 FROM airline_info
158 WHERE airline_country IN ('France', 'Portugal', 'Poland')
159 AND created_at BETWEEN '2023-11-01' AND '2024-03-31';
160
161
162 SELECT * FROM baggage
163 WHERE weight_in_kg > 25
164 ORDER BY weight_in_kg DESC
165 LIMIT 3;
166
167 SELECT baggage_id, weight_in_kg
168 FROM baggage
169 ORDER BY weight_in_kg DESC;
170
171 SELECT first_name, last_name
172 FROM passengers
173 ORDER BY date_of_birth DESC
174 LIMIT 1;

```

first_name	last_name
Timur	Kim

Task11:

The screenshot shows a PostgreSQL IDE interface. On the left, the 'Object Explorer' pane displays a tree structure of database objects. The 'baggage' table is selected, showing its columns: baggage_id (integer), weight_in_kg (numeric(4,2)), created_at (timestamp without time zone), updated_at (timestamp without time zone), and booking_id (integer). The main query editor displays a SQL query with line numbers 162 to 179. The query consists of two parts: a query on the 'baggage' table and a query on the 'booking' table. The results pane at the bottom shows the output of the second query, displaying the booking platform and the cheapest price for each platform.

```

162 SELECT * FROM baggage
163 WHERE weight_in_kg > 25
164 ORDER BY weight_in_kg DESC
165 LIMIT 3;
166
167 SELECT baggage_id, weight_in_kg
168 FROM baggage
169 ORDER BY weight_in_kg DESC;
170
171 SELECT first_name, last_name
172 FROM passengers
173 ORDER BY date_of_birth DESC
174 LIMIT 1;
175
176
177 SELECT booking_platform, MIN(ticket_price) AS cheapest_price
178 FROM booking
179 GROUP BY booking_platform;

```

booking_platform	cheapest_price
Airport Desk	15652.65
Mobile App	20863.30
Website	14404.90

Task12:

The screenshot shows a PostgreSQL IDE interface. The query editor contains the following SQL code:

```

166
167 SELECT baggage_id, weight_in_kg
168 FROM baggage
169 ORDER BY weight_in_kg DESC;
170
171 SELECT first_name, last_name
172 FROM passengers
173 ORDER BY date_of_birth DESC
174 LIMIT 1;
175
176
177 SELECT booking_platform, MIN(ticket_price) AS cheapest_price
178 FROM booking
179 GROUP BY booking_platform;
180
181
182 SELECT * FROM airline_info
183 WHERE airline_code ~ '[0-9]';

```

The Data Output tab shows the results of the last query (line 183):

airline_id [PK] integer	airline_code character varying (30)	airline_name character varying (50)	airline_country character varying (50)	created_at timestamp without time zone	updated_at timestamp without time zone
1	2	AL2	UK	2026-09-29 20:19:36.119471	2026-09-29 20:19:36.119471
2	3	AL3	JAPAN	2026-09-29 20:19:36.119471	2026-09-29 20:19:36.119471
3	4	AL4	FRANCE	2026-09-29 20:19:36.119471	2026-09-29 20:19:36.119471
4	6	AL6	FRANCE	2026-09-29 20:19:36.119471	2026-09-29 20:19:36.119471
5	7	AL7	USA	2026-09-29 20:19:36.119471	2026-09-29 20:19:36.119471
6	8	AL8	JAPAN	2026-09-29 20:19:36.119471	2026-09-29 20:19:36.119471
7	9	AL9	KAZAKHSTAN	2026-09-29 20:19:36.119471	2026-09-29 20:19:36.119471
8	10	AL10	USA	2026-09-29 20:19:36.119471	2026-09-29 20:19:36.119471
9	11	AL11	FRANCE	2026-09-29 20:19:36.119471	2026-09-29 20:19:36.119471
10	12	AL12	UK	2026-09-29 20:19:36.119471	2026-09-29 20:19:36.119471
11	13	AL13	JAPAN	2026-09-29 20:19:36.119471	2026-09-29 20:19:36.119471
12	14	AL14	USA	2026-09-29 20:19:36.119471	2026-09-29 20:19:36.119471

Summary: Total rows: 20. Query complete 00:00:00.071. Status: Successfully run. Total query runtime: 71 msec. 20 row.

Task13:

The screenshot shows a PostgreSQL IDE interface. The Object Explorer on the left shows the database structure, including tables like `airline`, `baggage`, `passengers`, and `booking`. The query editor contains the following SQL code:

```

171 SELECT first_name, last_name
172 FROM passengers
173 ORDER BY date_of_birth DESC
174 LIMIT 1;
175
176
177 SELECT booking_platform, MIN(ticket_price) AS cheapest_price
178 FROM booking
179 GROUP BY booking_platform;
180
181
182 SELECT * FROM airline_info
183 WHERE airline_code ~ '[0-9]';
184
185
186 SELECT * FROM airline_info
187 ORDER BY created_at DESC
188 LIMIT 5;

```

The Data Output tab shows the results of the last query (line 188):

airline_id [PK] integer	airline_code character varying (30)	airline_name character varying (50)	airline_country character varying (50)	created_at timestamp without time zone	updated_at timestamp without time zone
1	1	A1	AIRLINE_1	2026-10-06 12:12:08.585689	2026-10-06 12:12:08.585689
2	25	KAZ	AIRASTANA	2026-09-30 11:40:02.321177	2026-09-30 11:40:02.321177
3	24	FF	FLYFLY	2026-09-29 21:02:56.669773	2026-09-29 21:02:56.669773
4	22	AE	AIREASY	2026-09-29 21:02:56.669773	2026-09-29 21:02:56.669773
5	23	FH	FLYHIGH	2026-09-29 21:02:56.669773	2026-09-29 21:02:56.669773

Task14:

The screenshot shows the DBeaver interface with the following SQL queries in the editor:

```
176
177 SELECT booking_platform, MIN(ticket_price) AS cheapest_price
178 FROM booking
179 GROUP BY booking_platform;
180
181
182 SELECT * FROM airline_info
183 WHERE airline_code ~ '[0-9]';
184
185
186 SELECT * FROM airline_info
187 ORDER BY created_at DESC
188 LIMIT 5;
189
190
191 SELECT * FROM baggage_check
192 WHERE booking_id BETWEEN 200 AND 300
193 AND check_result <> 'Checked';
```

The Data Output tab shows the following columns:

baggage_check_id	check_result	created_at	updated_at	booking_id	passenger_id
[PK] integer	character varying (50)	timestamp without time zone	timestamp without time zone	integer	integer

Task15:

The screenshot shows the DBeaver interface with the following SQL queries in the editor:

```
181
182 SELECT * FROM airline_info
183 WHERE airline_code ~ '[0-9]';
184
185
186 SELECT * FROM airline_info
187 ORDER BY created_at DESC
188 LIMIT 5;
189
190
191 SELECT * FROM baggage_check
192 WHERE booking_id BETWEEN 200 AND 300
193 AND check_result <> 'Checked';
194
195
196 SELECT * FROM baggage_check
197 WHERE DATE_TRUNC('month', updated_at) = DATE_TRUNC('month', created_at)
198 AND updated_at < created_at;
```

The Data Output tab shows the following columns:

baggage_check_id	check_result	created_at	updated_at	booking_id	passenger_id
[PK] integer	character varying (50)	timestamp without time zone	timestamp without time zone	integer	integer